

Nihar Gaddam

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Computer Engineering – Applied Advanced Study

Dec 2023

- GPA: 3.75/4.00
- Relevant Coursework: Distributed Systems, Cloud Computing, Foundations of Computer Systems, Computer Vision, Machine Learning for Signal Processing, Introduction to Deep Learning

Sreenidhi Institute of Science and Technology

Hyderabad, India

Bachelor of Technology in Electronics and Communication Engineering

June 2020

- GPA: 9.39/10.00
- Minor in Computer Science and Engineering

SKILLS

Languages: Go, Rust, Python, C++, Java

IAAS: AWS, Azure, GCP

Container Platform: Docker, Kubernetes

CI/CD: GitHub Actions, Apache Airflow

IAC & Config Management: Terraform, Puppet

Storage: HBase, HDFS, Zookeeper, MySQL, MongoDB, Neo4j, Postgres

Frameworks/Tools: PyTorch, Spark, Kafka, Vertex AI

Monitoring and Observability: Prometheus, Grafana, Splunk, OpenTelemetry, SignalFx

EXPERIENCE

Cisco Systems Inc.

Pittsburgh, PA

Software Development Engineer

Aug 2024 – Present

- **Backend Engineering & API Development**
 - Designed and maintained features for a Go-based Kubernetes operator (control plane) managing the full lifecycle of tens of thousands of isolated, multi-tenant cloud environments, plus a customer-facing configuration API.
- **System Monitoring & Performance Optimization**
 - Monitored and optimized Kubernetes cluster deployments to resolve performance bottlenecks, reducing technical debt through regular refactors across development, staging, and production.
- **Key Projects and Achievements**
 - Replaced a months-long manual provisioning process with a declarative, self-serve workflow; scaled to ~100 environments/month with zero manual coordination after authoring the design proposal and driving adoption across five teams.
 - Built the platform's first metrics + SLO observability layer (Prometheus, OpenTelemetry), shipping automated detectors that catch failures proactively instead of via customer reports.
 - Diagnosed and fixed a disaster-recovery teardown ordering bug that could destroy a live primary environment while its replica was still active; redesigned the control loop to enforce dependency-safe teardown, preventing data loss.
 - Built an automated garbage-collection system for stale infrastructure, reclaiming 2,385 abandoned environments and cutting unnecessary cloud spend fleet-wide.
 - Automated service-to-service authentication with ephemeral credentials, removing 140+ manual tickets/year, and drove encryption-at-rest to 100% coverage across 3,000+ environments.
 - Enabled customer read access to standby replicas during regional failover, cutting recovery time 30% for 800+ customers.
 - Closed a network security-group filtering gap, reducing related incidents 24%, and surfaced certificate-rotation status to customers, cutting support tickets 23% for 8,000+ users.
 - Replaced ad-hoc cloud subscription-pool logic with a dedicated reconciliation service, making acquisition failures observable and retryable as native infrastructure conditions.
 - Automated authentication-token rotation across 15 service accounts, eliminating manual intervention and improving token-management efficiency ~40%.

Accenture Solutions Pvt. Ltd.

Hyderabad, India

Application Development Associate

Dec 2020 – Oct 2021

- **Cloud Infrastructure Management**
 - Migrated client infrastructure from on-premises to AWS and Azure, improving scalability and reducing operational costs by 20%; maintained development and production environments, reducing downtime by up to 40%.
 - Executed high-risk production changes with minimal disruption, maintaining a 99.8% success rate in change management.

- **Data Analysis and Reporting**

- Leveraged Power BI for real-time server performance trend analysis, reducing response time to performance issues by 30%; managed ETL dataflows from Azure SQL, SharePoint, Redshift, MongoDB, and Snowflake into BI reporting pipelines.

PROJECTS

Personal

timberdb  | *Go*

Jun 2026

- Built a time-partitioned, TTL-native LSM storage engine in Go; benchmarked 2.0× Pebble, 5.8× Badger, and 13× bbolt append throughput at 3 allocs/op, with 6.6× faster scans than Badger.

SVID-Exchange  | *Go*

Mar 2026

- Built a Zero Trust token exchange service that converts SPIFFE SVIDs into scoped, short-lived ES256 JWTs over mTLS, eliminating static shared secrets for service-to-service authentication.

Saga-Conductor  | *Go*

Mar 2026

- Built a lightweight saga-pattern orchestrator for distributed transactions: callers define each step with a forward action and a compensating action, and the conductor executes them in order, automatically rolling back completed steps in reverse on failure.

Academic

Distributed Backend Storage for Multiplayer Environments | *Go*

Dec 2023

- Created a distributed key-value storage system utilizing replication, partitioning, and concurrency mechanisms to support a globally distributed multiplayer online game, ensuring data consistency and low-latency access across multiple backend servers.

Analysis of Twitter User Graph with Spark | *Scala*

Nov 2023

- Proficiently applied Apache Spark programs on Azure HDInsight to analyze the Twitter social graph; implemented the PageRank algorithm to identify influential users and optimized runtime from 60 to 28 minutes, enhancing performance.

Advanced Resource Scaling using Cloud SDKs | *Python*

Sep 2023

- Implemented horizontal scaling with the Boto3 AWS SDK, reducing response times from 30 to 8 minutes; increased throughput from 8K to 40K RPS; configured Elastic Load Balancers, Auto Scaling Groups, and Terraform for efficient deployment.

CERTIFICATIONS

Certified Kubernetes Application Developer (CKAD)  — Cloud Native Computing Foundation (CNCF) 2024